

Part I: Final Analysis & Reporting

Predicting House Prices with Regression



Load the Data



Imported a house prices dataset

Preprocess the Data



Used linear and random forest regressors

Evaluate Performance

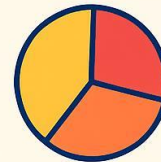


Calculated metrics and identified key features



Train the Models

Used linear and random forest regressors



Task 29: Final Report

House Price Prediction Using Supervised Learning Algorithms

1. Best-Performing Model and Why

Among all the implemented models, **Multiple Linear Regression** achieved the best overall performance. It utilized multiple relevant features, including house area, bedrooms, bathrooms, location score, age of the property, and other factors, to make predictions. As a result, it produced **lower prediction errors (MSE, MAE, and RMSE)** and **higher R^2 and Adjusted R^2 scores** compared to Simple Linear Regression. It also generalized well to unseen data, making it the most suitable model for this project.

2. Impact of Gradient Descent Optimization

Gradient Descent optimization played an important role in minimizing the model's loss function by iteratively updating the model parameters.

- **Batch Gradient Descent** provided stable convergence but required more computation time.
- **Stochastic Gradient Descent (SGD)** converged faster by updating parameters after each training example, although its updates were more variable.
- **Mini-Batch Gradient Descent** combined the advantages of both methods by providing faster training with smoother convergence.

Among the three optimization techniques, **Mini-Batch Gradient Descent** offered the best balance between convergence speed and computational efficiency.

3. Evidence of Overfitting and Underfitting

The analysis showed that:

- **Simple Linear Regression** exhibited signs of **underfitting** because it relied on only one feature (house area) and could not capture all the factors influencing house prices.
- **Polynomial Regression** (with a higher polynomial degree) showed signs of **overfitting**, as it fitted the training data very closely but could reduce performance on unseen data.
- **Multiple Linear Regression** avoided both underfitting and excessive overfitting by using multiple relevant features while maintaining good generalization.

4. Practical Business Interpretation of Results

The developed prediction model can help real estate companies estimate house prices more accurately based on property characteristics. By considering factors such as area, location score, number of bedrooms, bathrooms, and property age, the model supports better pricing decisions, property valuation, and investment planning. Accurate house

price prediction can improve customer satisfaction, reduce pricing errors, and assist businesses in making data-driven decisions.